

GIRRAWEE HIGH SCHOOL

MATHEMATICS

Year 11
Task 4

14th August 2006
Time allowed 90 minutes

INSTRUCTIONS : Attempt all 6 questions..

Write your answers on your own paper.

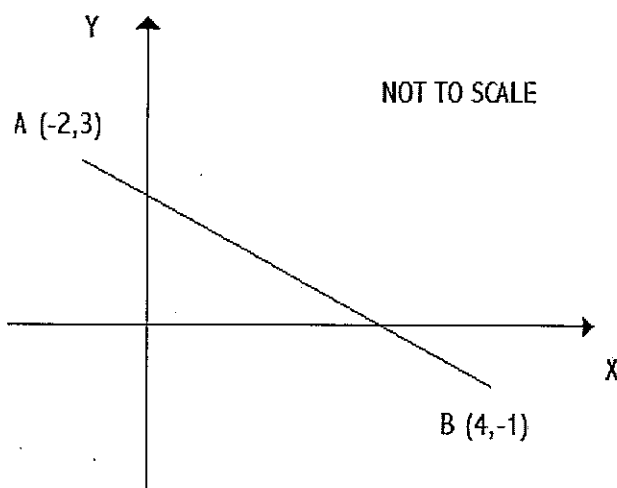
All necessary working must be shown.

Marks may be deducted for careless or poorly arranged work.

Begin each question on a new page.

Question 1. (11 Marks)

Marks



In the diagram above, the interval AB joining the points A (-2, 3) and B (4, -1) is shown. Find

- | | |
|--|---|
| (a) The gradient of the line AB. | 2 |
| (b) The equation of the line AB | 2 |
| (c) The distance AB (leave your answer as a surd) | 2 |
| (d) The mid point M of AB | 1 |
| (e) The equation of the line through M perpendicular to AB | 2 |
| (f) Find the angle that the line CM makes with the x axis (nearest degree) | 2 |

Question 2 (14 Marks)

- | | |
|---|---|
| (a) Find the perpendicular distance from the point (-1, 5) to the line $3x - 4y - 12 = 0$ | 3 |
| (b) Find the point of intersection of the lines $3x + y - 1 = 0$ and $2x + 3y + 4 = 0$ | 3 |
| (c) Find the point of intersection of the circle $(x-1)^2 + y^2 = 2$ and the line $x + y + 1 = 0$ | 4 |
| (d) Find the equation of the line parallel to the line $3x - y - 3 = 0$ that passes through the point of intersection of $3x + y - 1 = 0$ and $x - y + 4 = 0$ | 4 |

Question 3 (11 Marks)

Marks

(a) Evaluate the following limits

(i) $\lim_{x \rightarrow 3} \frac{2x-6}{x}$

(ii) $\lim_{x \rightarrow 2} \frac{x^2-4}{x-2}$

3

(iii) $\lim_{x \rightarrow \infty} \frac{2x^2-3x+7}{x^2-2}$

(iv) $\lim_{x \rightarrow 3} \frac{x^2-5x+6}{x-3}$

4

(b) If $f(x) = x^2 + 2x$ differentiate $f(x)$ from first principles

4

Question 4 (15 Marks)

Differentiate the following functions

(i) $y = 3x$

(ii) $y = x^2$

3

(iii) $y = \frac{1}{x}$

(iv) $y = \sqrt{2x}$

4

(v) $y = 3x^4 - 5x + 7$

(vi) $y = 3x^{\frac{3}{2}} - 5x^{\frac{7}{3}}$

4

(vii) $y = \frac{3}{\sqrt{x}} + x^5$

(viii) $y = \frac{7}{x^3} + \sqrt[3]{x}$

4

Question 5 (21 Marks)

(a) Differentiate the following

(i) $y = (x^3 + 3x^2 - 9)^{10}$

(ii) $y = \frac{3x^2 - 5x + 7}{2x^2}$

6

(iii) $y = (4x^2 - 7)(5x + 4)^3$

(iv) $y = \frac{1}{\sqrt{x^3 - 7x}}$

7

(v) $y = \frac{(5-x)^3}{x^2-3}$

4

(b) If $f(x) = 3x^2 - 4x + 5$ evaluate

(i) $f'(x)$

(ii) $f'(0)$

2

(iii) $f'(2)$

(iv) Solve for x ; $f'(x) = 0$

2

Question 6 (14 Marks)

Marks

(a) If $y = 2x^2 - 3x + 5$

(i) Find the equation of the tangent to the curve at the point $(1, 4)$. 3

(ii) Find the equation of the normal to the curve at the point $(1, 4)$ 2

(b) Find the point on the curve $y = x\sqrt{x}$ where the tangent is parallel to the line $2x - 3y + 5 = 0$ 4

(c) Find the equations of the tangents to the curve $y = x^3 - 3x^2 + 2x - 4$ that are parallel to the line $y - 2x - 7 = 0$ 5

YEAR 11 MATHEMATICS

TASK 4 2006
Question 1
a) $M_{AB} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3+1}{-2-4} = -\frac{2}{3}$ (2)

b) EQⁿ AB
 $y - y_1 = m(x - x_1)$
 $y + 1 = -\frac{2}{3}(x - 4)$
 $y = -\frac{2}{3}x + \frac{5}{3}$
OR $2x + 3y - 5 = 0$ (2)

c) $D_{AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
 $= \sqrt{(-2 - 4)^2 + (3 + 1)^2}$
 $= \sqrt{32}$ OR $2\sqrt{2}$ (2)

d) MID PT = $M(\frac{-2+4}{2}, \frac{3+1}{2})$
 $= M(1, 1)$ M

e) $m_1 \times m_2 = -1$ $m_2 = \frac{3}{2}$
 $y - 1 = \frac{3}{2}(x - 1)$
 $y = \frac{3}{2}x - \frac{1}{2}$
OR $3x - 2y - 1 = 0$ (2)

f) $\tan \theta = m$
 $\tan \theta = -\frac{2}{3}$ $\theta = -34^\circ$
 $\theta = 146^\circ$ OR 34° (2)

Question 2

a) $D_{PERP} = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$
 $= \frac{|3(-1) + 4(5) - 12|}{\sqrt{3^2 + 4^2}}$
 $= \frac{35}{5} = 7$ (3)

b) $3x + y = 1$
 $2x + 3y = -4$

$-9x + 3y = 3$ (A)
 $2x + 3y = -4$ (B)
 $7x = 7 \therefore x = 1$ $y = -2$
(1, -2) (3)

c) $(x-1)^2 + y^2 = 2$ (A)
 $x + y = -1$ (B)

From (B) $y = -(x+1)$
Subst (A) $(x-1)^2 + (-(x+1))^2 = 2$

$x^2 - 2x + 1 + x^2 + 2x + 1 = 2$

$2x^2 + 2 = 2$

$2x^2 = 0$

$x = 0$ $y = -1$ (0, -1) (4)

d) $3x + y = 1$ (A)
 $x - y = -4$ (B)

$4x = -3$ (A) + (B)

$x = -\frac{3}{4}$ $y = \frac{13}{4}$

ft to $3x - y - 3 = 0$
 $y = 3x - 3$
gradient is 3.

EQⁿ

$y - \frac{13}{4} = 3(x + \frac{3}{4})$

$y = 3x + \frac{11}{2}$

OR $6x - 2y + 11 = 0$ (4)

Question 3.

i) $\lim_{x \rightarrow 3} \frac{2x-6}{3} = \frac{0}{3} = 0$ (1)

ii) $\lim_{x \rightarrow 2} \frac{x^2-4}{x-2} = \lim_{x \rightarrow 2} \frac{(x+2)(x-2)}{(x-2)}$

$= \lim_{x \rightarrow 2} x+2$

$= 4$ (2)

iii) $\lim_{x \rightarrow \infty} \frac{2x^2-3x-7}{x^2-2}$

$= \lim_{x \rightarrow \infty} \frac{2 - \frac{3}{x} - \frac{7}{x^2}}{1 - \frac{2}{x^2}}$

$= \lim_{x \rightarrow \infty} \frac{2 - 0 - 0}{1 - 0}$

$= 2$ (2)

iv) $\lim_{x \rightarrow 3} \frac{x^2-5x+6}{x-3}$

$= \lim_{x \rightarrow 3} \frac{(x-3)(x-2)}{x-3}$

$= \lim_{x \rightarrow 3} x-2 = 1$ (2)

b) $f(x) = x^2 + 2x$

$f(x+h) = (x+h)^2 + 2(x+h)$

$= x^2 + 2xh + h^2 + 2x + 2h$

$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 2x + 2h - (x^2 + 2x)}{h}$

$= \lim_{h \rightarrow 0} \frac{2xh + h^2 + 2h}{h}$

$= \lim_{h \rightarrow 0} 2x + 2 + h$

$= 2x + 2$ (4)

Question 4

i) $y = 3x$

$\frac{dy}{dx} = 3$ (1)

ii) $y = x^2$

$\frac{dy}{dx} = 2x$ (2)

iii) $y = \frac{1}{x}$

$y = x^{-1}$

$\frac{dy}{dx} = -1x^{-2}$
 $= -\frac{1}{x^2}$ (2)

iv) $y = \sqrt{2x}$

$y = (2x)^{1/2}$

$\frac{dy}{dx} = \frac{1}{2}(2x)^{-1/2} \cdot 2$

$= \frac{1}{\sqrt{2x}}$ (2)

v) $y = 3x^4 - 5x + 7$

$\frac{dy}{dx} = 12x^3 - 5$ (2)

vi) $y = 3x^{3/2} - 5x^{7/3}$

$\frac{dy}{dx} = \frac{9}{2}x^{1/2} - \frac{35}{3}x^{4/3}$ (2)

vii) $y = \frac{3}{\sqrt{2x}} + x^5$

$y = 3x^{-1/2} + x^5$

$\frac{dy}{dx} = -\frac{3}{2}x^{-3/2} + 5x^4$

OR $= -\frac{3}{2\sqrt{2x}} + 5x^4$ (2)

viii) $y = \frac{1}{x^3} + 3\sqrt{x}$

$y = x^{-3} + x^{1/3}$

$\frac{dy}{dx} = -21x^{-4} + \frac{1}{3}x^{-2/3}$

$= -\frac{21}{x^4} + \frac{1}{3\sqrt[3]{x^2}}$ (2)

Question 5.

i) $y = (x^3 + 3x^2 - 9)^{10}$

let $u = x^3 + 3x^2 - 9$

$\frac{du}{dx} = 3x^2 + 6x$

$\therefore y = u^{10}$

$\frac{dy}{du} = 10u^9$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$= 10u^9 \cdot (3x^2 + 6x)$

$= 10(3x^2 + 6x)(x^3 + 3x^2 - 9)^9$ (3)

ii) $y = \frac{3x^2 - 5x + 7}{2x^2}$

$y = 3 - 5x^{-1} + 7x^{-2}$

$\frac{dy}{dx} = 5x^{-2} + 14x^{-3}$

$= \frac{5}{x^2} - \frac{14}{x^3}$ (3)

iii) $y = (4x^2 - 7)(5x + 4)^3$

$\frac{du \cdot v}{dx} = v \cdot \frac{du}{dx} + u \cdot \frac{dv}{dx}$

let $u = 4x^2 - 7$

$\frac{du}{dx} = 8x$

let $v = (5x + 4)^3$

$\frac{dv}{dx} = 15(5x + 4)^2$

$\frac{du \cdot v}{dx} = (5x + 4)^3 \cdot 8x + (4x^2 - 7) \cdot 15(5x + 4)^2$

$= (5x + 4)^2 \{ (5x + 4)8x + 15(4x^2 - 7) \}$

$= (5x + 4)^2 (100x^2 + 32x - 105)$ (4)

(iv) $y = \frac{1}{\sqrt{x^3 - 7x}}$

$y = (x^3 - 7x)^{-1/2}$

let $u = x^3 - 7x$

$\frac{du}{dx} = 3x^2 - 7$

$y = u^{-1/2}$

$\frac{dy}{du} = -1/2 u^{-3/2}$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$= \frac{-1}{2 u^{3/2}} \cdot (3x^2 - 7)$

$= \frac{7 - 3x^2}{2 (x^3 - 7x)^{3/2}}$ (3)

(v) $y = \frac{(5-x)^3}{x^2 - 3}$

let $u = (5-x)^3$

$\frac{du}{dx} = 3(5-x)^2 (-1)$

$v = x^2 - 3$

$\frac{dv}{dx} = 2x$

$\frac{du \cdot v}{dx} = v \cdot \frac{du}{dx} - u \cdot \frac{dv}{dx}$

$= \frac{(x^2 - 3)(-3)(5-x)^2 - (5-x)^3 \cdot 2x}{(x^2 - 3)^2}$

Q > con

$= - (5-x)^2 \left\{ \frac{3x^2 - 9 + 10x - 2x^2}{(x^2 - 3)^2} \right\}$

$= - (5-x)^2 \left\{ \frac{x^2 + 10x - 9}{(x^2 - 3)^2} \right\}$ (4)

b) $f(x) = 3x^2 - 4x + 5$

(i) $f'(x) = 6x - 4$ (1)

(ii) $f'(6) = -4$ (1)

(iii) $f'(2) = 12 - 4 = 8$ (1)

(iv) $f'(x) = 0$
 $0 = 6x - 4$
 $6x = 4$
 $x = 2/3$ (1)

Question 6

(a) (i) $y = 2x^2 - 3x + 5$

$\frac{dy}{dx} = 4x - 3$

$\frac{dy}{dx} (x=1) = 4 - 3 = 1$

$\therefore$ EQN OF TANGENT

$y - 4 = 1(x - 1)$

$y = x + 3$ (3)

(ii) EQN OF NORMAL

$y - 4 = -1(x - 1)$

$y = -x + 5$ (2)

(b) $y = x\sqrt{x}$
 $y = x^{3/2}$
 $\frac{dy}{dx} = 3/2 x^{1/2}$

NOW $2x - 3y + 5 = 0$ (4)

$3y = 2x + 5$

$y = 2/3 x + 5/3$

$\therefore$ gradient $= 2/3$

$\therefore 3/2 \sqrt{x} = 2/3$

$\sqrt{x} = 4/9$

$x = 16/81$

Pt $(16/81, 64/729)$ (4)

(4) $y = x^3 - 3x^2 + 2x - 4$

$\frac{dy}{dx} = 3x^2 - 6x + 2$

NOW $y = 2x + 7$
HAS GRADIENT 2

$\therefore 2 = 3x^2 - 6x + 2$

$0 = 3x^2 - 6x$

$0 = 3x(x - 2)$

$\therefore x = 0, 2$

$y = -4, -4$

TANG 1

$y + 4 = 2(x - 0)$

$y = 2x - 4$

TANG 2

$y + 4 = 2(x - 2)$

$y = 2x - 8$ (5)